

Project Title: VirtualFit: AI-Powered Virtual Try-On System

Course: Computer Vision (Final Project) Team Members:

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1. Abstract The rapid growth of e-commerce has revolutionized the fashion industry, yet the inability to physically try on clothes remains a significant challenge, leading to high return rates and customer dissatisfaction. This project introduces **VirtualFit**, a virtual try-on application powered by the **IDM-VTON** (Improving Diffusion Models for Virtual Try-On) architecture. Our system allows users to upload an image of themselves and a target garment to generate a photorealistic visualization of the clothing on their body. By integrating advanced diffusion models with a user-friendly web interface, VirtualFit provides a seamless and realistic fitting experience.

2. Introduction Online shopping offers convenience, but the lack of physical interaction with products often results in uncertainty regarding fit and style. Traditional virtual try-on solutions typically rely on 3D mesh warping (e.g., TPS), which often produces unnatural distortions and fails to preserve garment textures. To address this, we implemented a diffusion-based approach that not only fits the clothing to the user's body shape but also maintains high-fidelity details such as logos, patterns, and folds.

3. Methodology 3.1 Model Architecture We utilized the **IDM-VTON** model, which improves upon standard diffusion inpainting by incorporating two specialized modules:

- IP-Adapter:** Extracts high-level semantic features from the garment image (CLIP image embeddings) to ensure the generated clothing matches the reference.
- GarmentNet:** Encodes low-level details (texture, text) to preserve the realistic look of the fabric.

3.2 Implementation Pipeline The system processes inputs in the following stages:

- Input:** User image and Garment image.
- Preprocessing:** Automatic human parsing (masking) to identify the upper body area.

3. **Inference:** The diffusion model generates the try-on result using the garment features and the masked body image.
4. **UI/UX:** A Gradio-based frontend hosted on Google Colab provides an interactive interface for users.

4. Experiments and Results We tested the system with various clothing types (t-shirts, long-sleeve shirts) and different human poses.

- **Qualitative Analysis:** The model successfully handles complex scenarios, such as tucked-in shirts and different arm poses. The texture preservation is significantly better than traditional GAN-based methods.
- **User Interface:** The web application supports drag-and-drop uploads and generates results in approximately 30-40 seconds on a T4 GPU.

[Insert your 'result_example.png' image here in Word] Figure 1: Comparison of original user image, target garment, and the final VirtualFit result.

5. Conclusion VirtualFit demonstrates the potential of generative AI in fashion e-commerce. By leveraging IDM-VTON, we achieved photorealistic virtual try-on results that outperform standard warping techniques. Future work could include real-time video try-on and support for full-body outfits.

6. References

1. Choi et al., "IDM-VTON: Improving Diffusion Models for Authentic Virtual Try-On in the Wild", 2024.
2. VITON-HD Dataset.